SYSTEM STATUS: OPERATIONAL

DATA STREAM: ACTIVE

Vizbin: Visual Reverse-Engineering for the Command Line

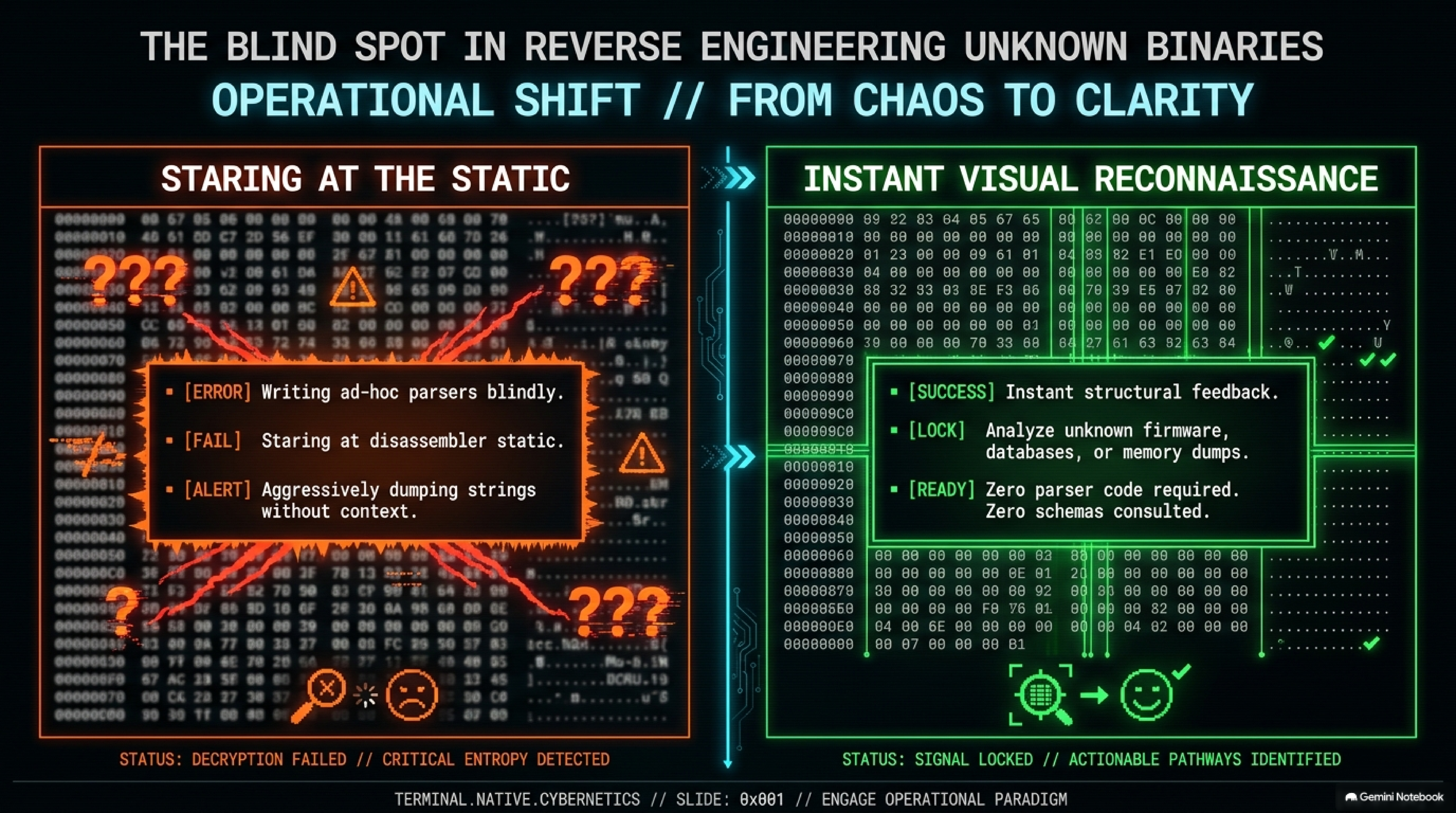
```
[user@vizbin ~]$

> Take a blob.
> Pretend it is an image.
> Vary the lie until the truth starts to show.
```

Zero-dependency visual reconnaissance for firmware, memory dumps, and packet captures.

SYSTEN STATUS: OPERATIONAL

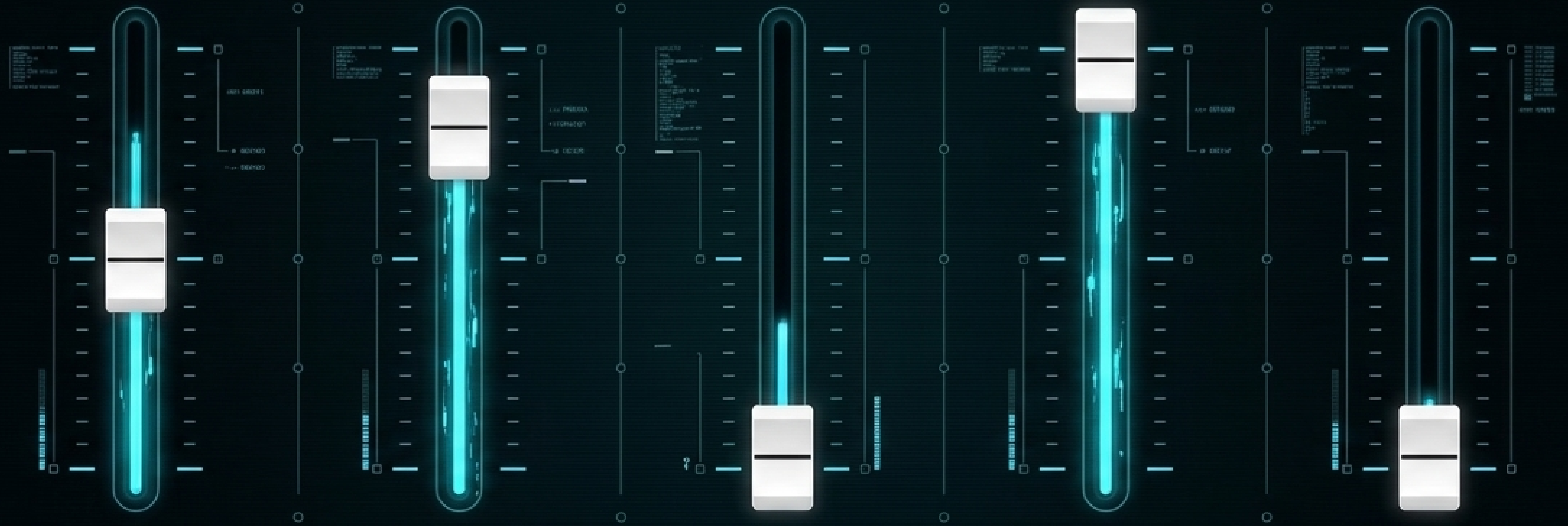
DATA STREAM: ACTIVE


THE BLIND SPOT IN REVERSE ENGINEERING UNKNOWN BINARIES
OPERATIONAL SHIFT // FROM CHAOS TO CLARITY
STARING AT THE STATIC
▪ [ERROR] Writing ad-hoc parsers blindly.
▪ [FAIL] Staring at disassembler static.
▪ [ALERT] Aggressively dumping strings without context.
???
???
???
?
STATUS: DECRYPTION FAILED // CRITICAL ENTROPY DETECTED
INSTANT VISUAL RECONNAISSANCE
▪ [SUCCESS] Instant structural feedback.
▪ [LOCK] Analyze unknown firmware, databases, or memory dumps.
▪ [READY] Zero parser code required. Zero schemas consulted.
STATUS: SIGNAL LOCKED // ACTIONABLE PATHWAYS IDENTIFIED
TERMINAL.NATIVE.CYBERNETICS // SLIDE: 0x001 // ENGAGE OPERATIONAL PARADIGM
Gemini Notebook

TREATING VISUAL RENDERING AS A SCIENTIFIC METHOD

ALMOST EVERYTHING YOU DO IN VIZBIN IS A HYPOTHESIS.

CONTROL PANEL



WIDTH
What if the data repeats every N bytes?
(Stride/Periodicity)

PROJECTION
What property of these bytes matters?
(Lenses)

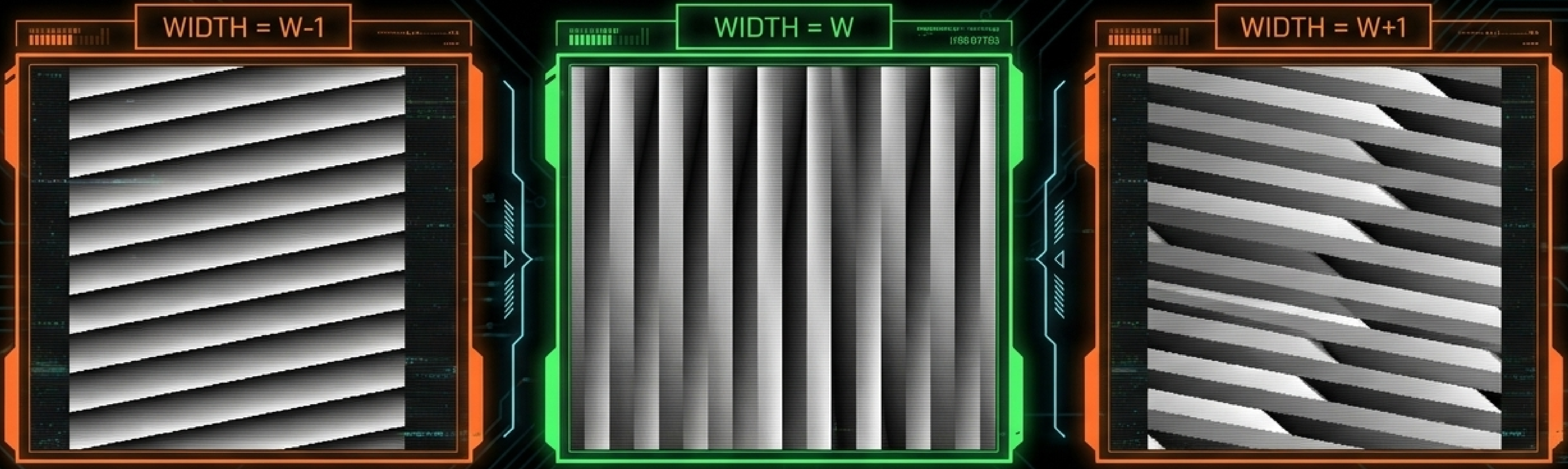
TRANSFORM CHAINS
What if I process that property further?
(Composition in Depth)

RGB COMPOSITION
What if I ask several questions at once?
(Composition in Breadth)

OFFSET
What if the structure actually starts here?
(Origin/Phase)

FINDING THE STRIDE LOCKS STRUCTURE INTO PLACE

V-HOLD EXPLANATORY DIAGRAM // SIGNAL ALIGNMENT PROTOCOL



SHEARED STRUCTURE | SIGNAL LOCK (188 BYTES) | SHEARED STRUCTURE

Visual stride spectroscopy: ranks adjacent-row coherence →

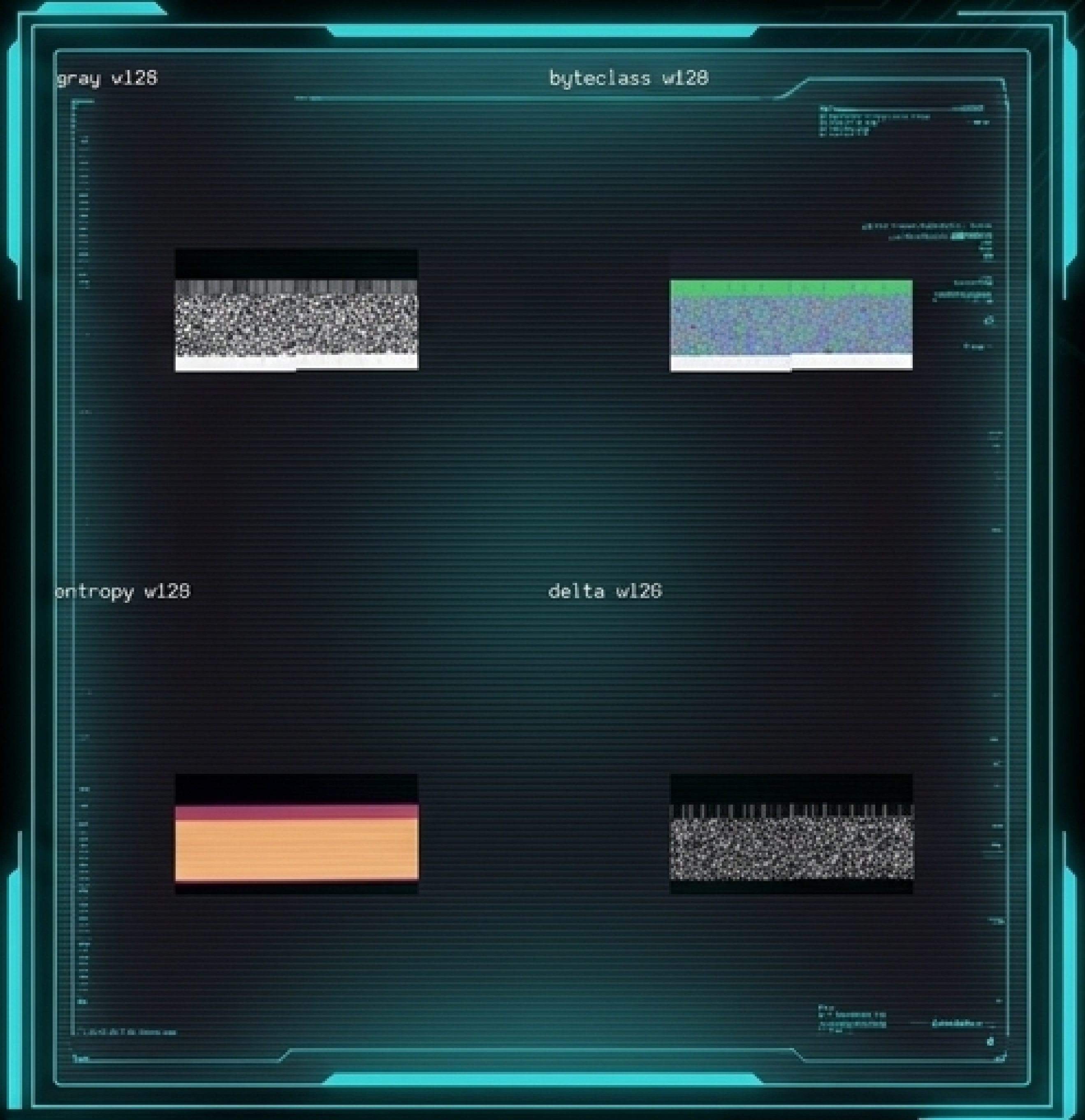
```
vizbin suggest foo.bin
vizbin animate foo.bin --widths 180,184,188,192,196 -m gray
```

← Sweeps widths dynamically, letting structure snap into alignment

LENSES THAT REVEAL DIFFERENT STRUCTURAL CHARACTERISTICS

LENS DIAGNOSTIC MATRIX

MODE	THE QUESTION	BEST FOR
`gray`	What are the byte values?	Raw periodicity, padding, fine texture.
`byteclass`	What kinds of bytes are here?	Separating nulls, printables, and high-bits.
`entropy`	How unpredictable is this region?	Transitions between code, padding, and compressed payloads.
`delta`	How fast do values change?	Exposing slow ramps and counters.
`text`	Where do the strings live?	A visual strings tool that prints readable ASCII glyphs in-line while keeping the surrounding binary scaffolding visible as structural tiles.



SENSOR READOUT: STRUCTURAL PROJECTIONS

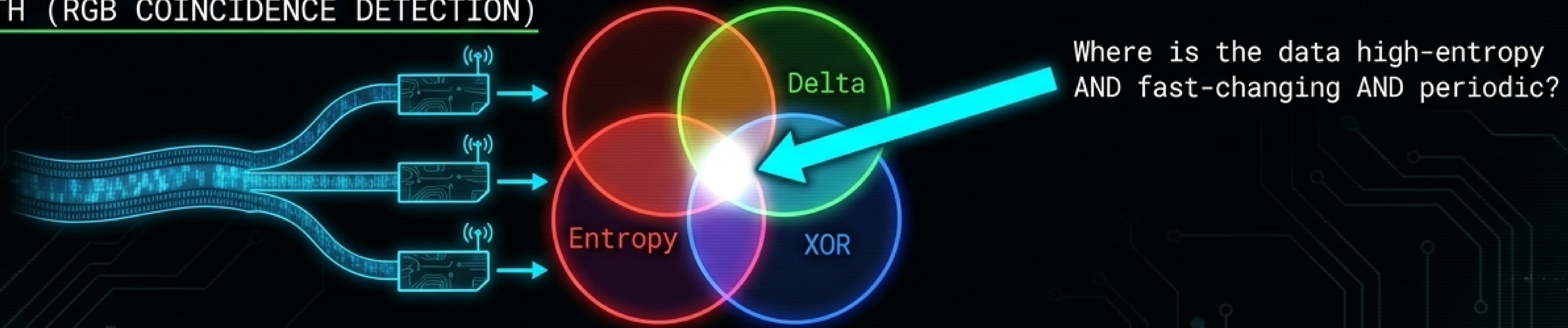
SCALING HYPOTHESES THROUGH DEPTH AND BREADTH

PIPELINE ARCHITECTURE DIAGRAM // TRANSFORM CHAINING & VISUAL COINCIDENCE

DEPTH (TRANSFORM CHAINING)

```
> vizbin -t xor,entropy
```

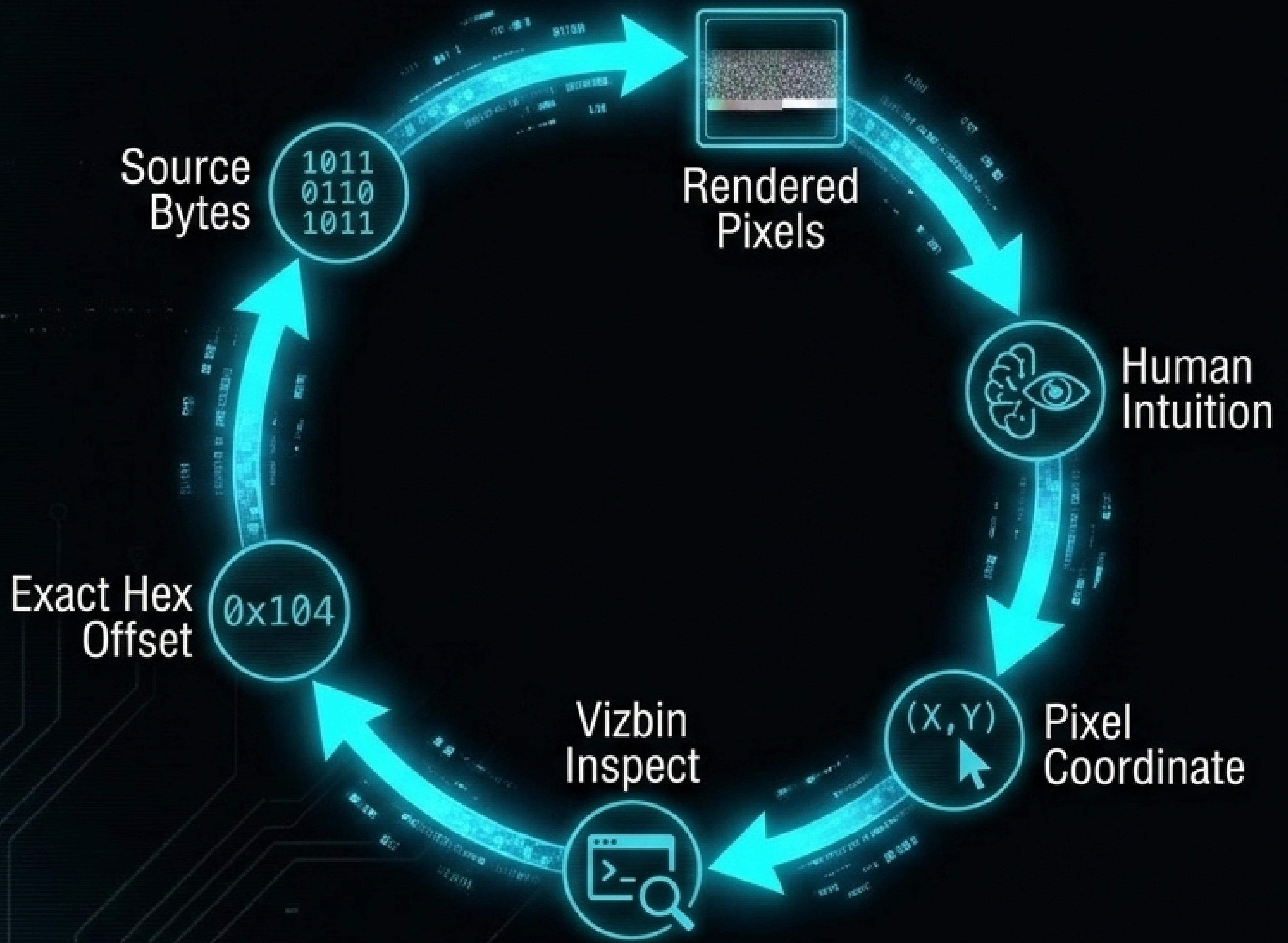
BREADTH (RGB COINCIDENCE DETECTION)




```
> vizbin --rgb entropy,delta,xor
```

BRIDGING HUMAN INTUITION BACK TO HARD MATHEMATICAL TRUTH

BIDIRECTIONAL ANALYTICAL BRIDGE // CYCLICAL FEEDBACK LOOP




```
> vizbin inspect foo.bin -w 128 --rgb
  entropy,delta,xor --offset 0x104

byte offset 0x104 -> pixel (4, 4)
R(entropy)=0x19  G(delta)=0x13  B(xor)=0x12
```

Unlike GUI hover tooltips, inspect is terminal-native, programmable, mode-aware, and provides multi-projection translations (XOR operands, local entropy, printable chars).

AN OPERATIONAL EDGE BUILT FOR HOSTILE ENVIRONMENTS

OPERATIONAL EDGE MATRIX // VIZBIN VS GUI VISUALIZERS

	Vizbin	GUI Visualizers (Binocle, Cantor.dust)
Instant Recon	✓	✓
Zero-Dependencies	✓	✗
Air-gap / SSH Viable	✓	✗
Terminal Programmability	✓	✗
Pipeline Composability	✓	✗

Written in pure Python standard library. Even the LZW GIF encoder and BMP writer are built from scratch. Drop the raw script on a forensics box over SSH with no internet, no X11, and no pip.

THE v0.5.0 ROADMAP: STRUCTURE INFERENCE AND SCALE

ROADMAP OVERVIEW // v0.5.0 OBJECTIVES

STRUCTURE INFERENCE (THE FLAGSHIP)

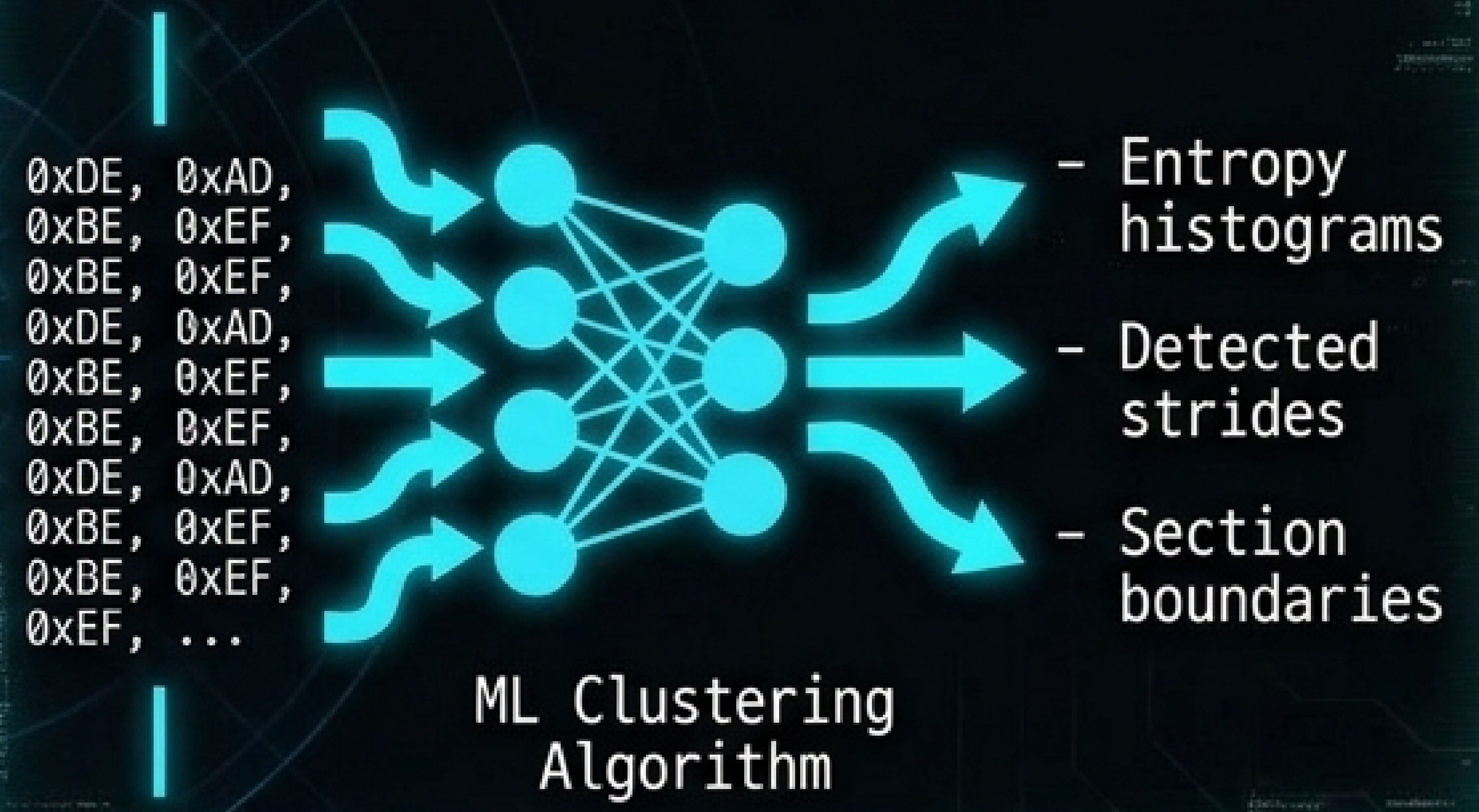
`suggest finds stride N`

Profiles column grid
(byte-class, entropy, delta)

Draft Layout: 188-byte records:
col 0 = counter, cols 12-31 = ASCII

Exports to Kaitai .ksy stubs, Python structs, or 010 Editor templates.
No other visualizer does this.

MACHINE-READABLE SENSOR (profile --json)



Enables terminal-pipe clustering of 100k malware or firmware samples.

The v0.5.0 Roadmap: Native RE and New Senses

ROADMAP OVERVIEW // v0.5.0 OBJECTIVES

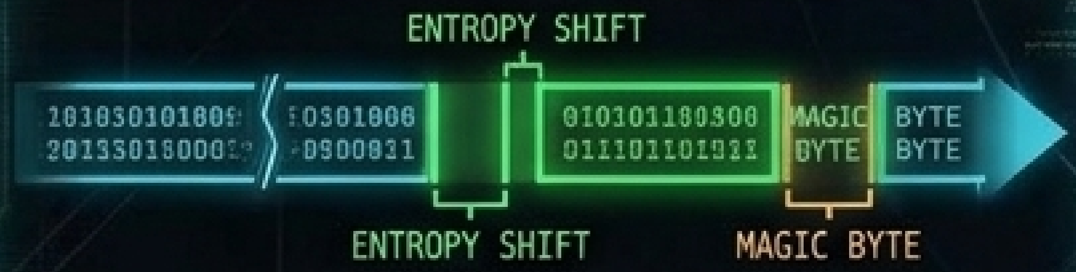
Structural Diff (diff)

Align and highlight visual/structural changes between firmware versions or malware variants.

Terminal Rendering (--term)

Low-friction 24-bit ANSI + half-block rendering directly into the SSH terminal window without X11 or file transfer.

Auto-Segmentation (segments)



Carving files by combining entropy transitions, magic bytes, and stride shifts.

Wildcard: Sonification

Mapping bytes/entropy to pure-stdlib WAV audio so periodicity becomes startlingly audible.

Zero-dependency deployment and essential commands

SCANNABLE GRID SHEET

Requirements

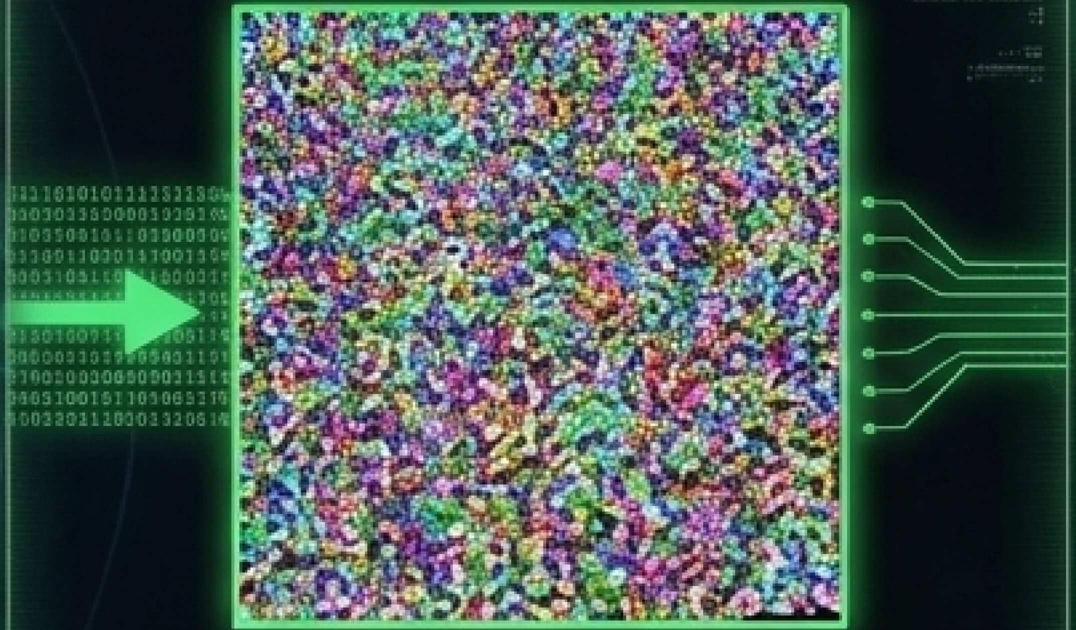
Python 3.9+. Zero dependencies.

```
>_ pip install vizbin (or run the raw script)
```

Command Grid

render :	render one image	Render one image
suggest:	candidate widths	Candidate widths
contact:	grid of widths/modes	Grid of widths/modes
animate:	sweep widths to GIF	Sweep widths to GIF
inspect:	pixel/offset mapping	Pixel/offset mapping

The Reversible Payload Trick



vizbin bmp and unbmp enable reversible payload-as-pixel round-trips.

Payload byte N lives exactly at BMP offset 54 + N. Identical recovery guaranteed.